LAB EXPERIMENT RECORD

15-10-2026---Set 19: Survey Responses Analysis

Department	AIML
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R and Tableau
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Date	15 OCTOBER 2026
Day	THURSDAY

Aim

To analyze survey response data using a grouped bar chart, stacked bar chart and table, and to build an interactive Tableau dashboard for exploring the survey responses.

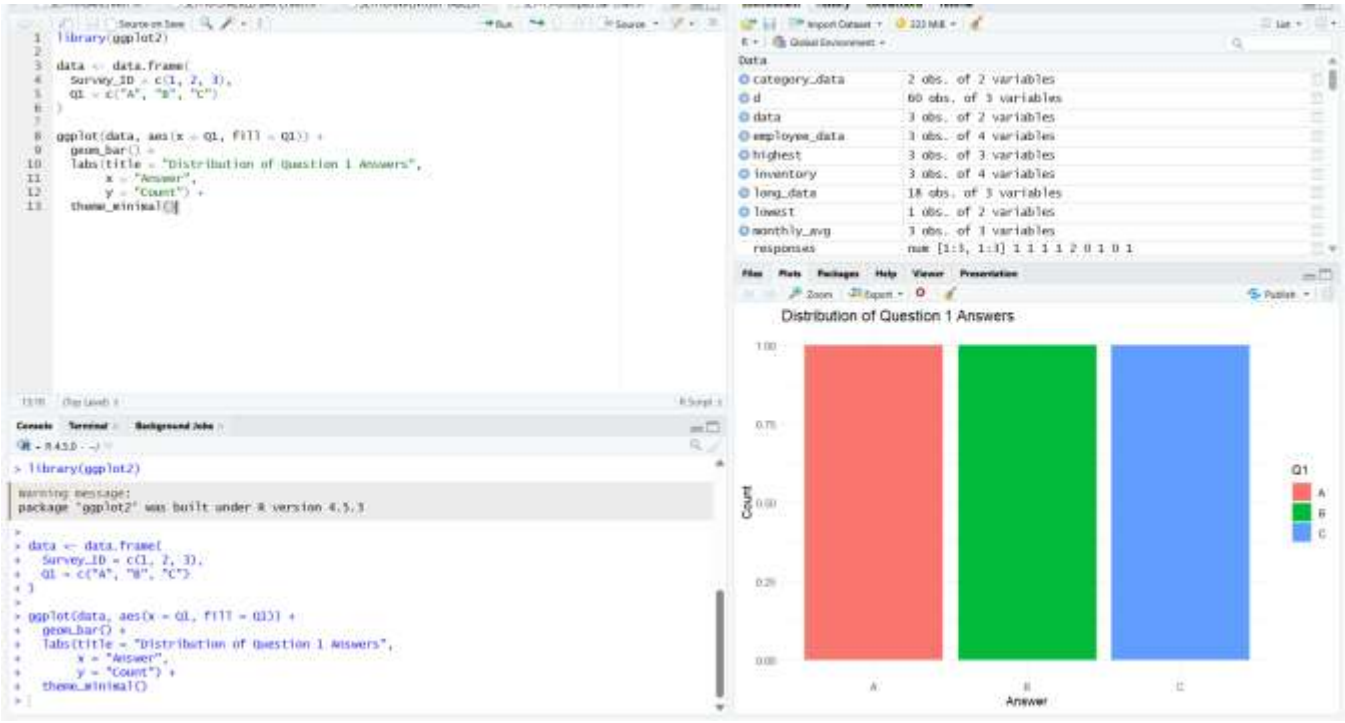
Dataset: Survey Responses

Survey ID	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B

Question 1: Grouped Bar Chart of Question 1 Answers

Question: Create a grouped bar chart to visualize the distribution of answers for Question 1. Label the chart elements.

Aim: To visualize how many respondents chose each answer for Question 1 using a labelled grouped bar chart.

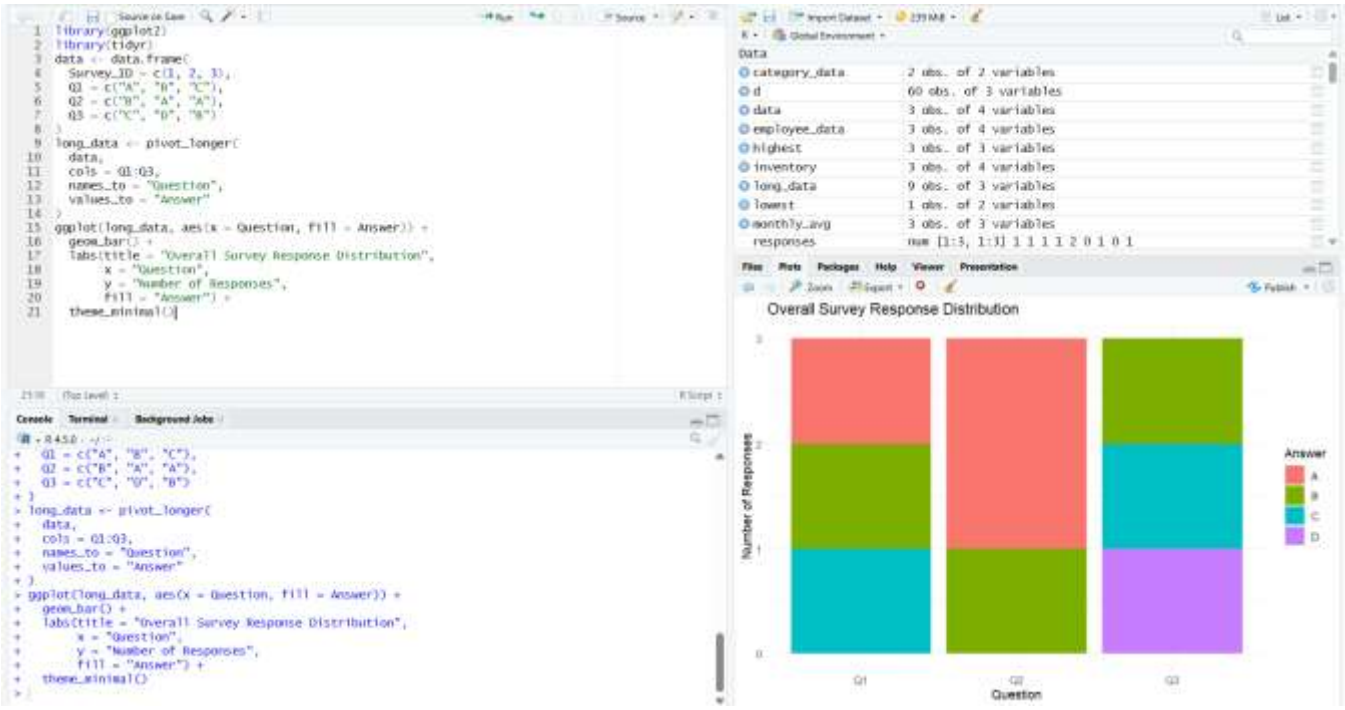


Result: Answers A, B and C were each chosen by 1 respondent and D by none, so the answers are evenly spread.

Question 2: Stacked Bar Chart of All Responses

Question: Generate a stacked bar chart to represent the overall distribution of responses for all three questions.

Aim: To represent the overall distribution of answers across Questions 1, 2 and 3 using a stacked bar chart.



Result: Answer A is chosen most in Question 2 (2 of 3 respondents), while Question 3 is the only question where D appears.

Question 3: Table of Survey Responses

Question: Build a table to show the survey response data for each survey. Label the table elements.

Aim: To present the answers given by each survey respondent in a clearly labelled table.

```
1  data <- data.frame(
2    Survey_ID = c(1, 2, 3),
3    Q1 = c("A", "B", "C"),
4    Q2 = c("B", "A", "A"),
5    Q3 = c("C", "D", "B")
6  )
7
8  print(data, row.names = FALSE)
```

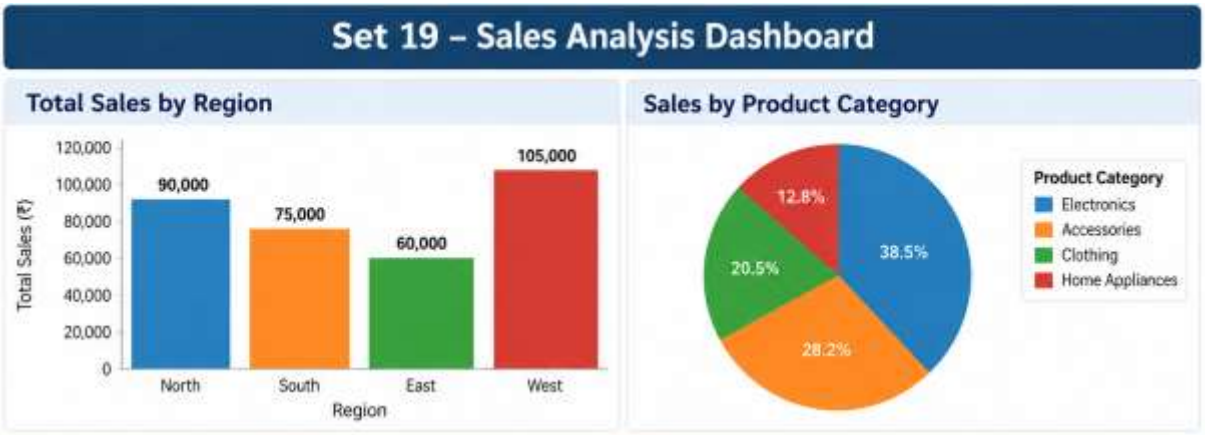
```
+   labs(title = "Overall Survey Response Distribution",
+        x = "Question",
+        y = "Number of Responses",
+        fill = "Answer") +
+   theme_minimal()
> data <- data.frame(
+   Survey_ID = c(1, 2, 3),
+   Q1 = c("A", "B", "C"),
+   Q2 = c("B", "A", "A"),
+   Q3 = c("C", "D", "B")
+ )
>
> print(data, row.names = FALSE)
 Survey_ID Q1 Q2 Q3
         1  A  B  C
         2  B  A  D
         3  C  A  B
>
```

Result: The table lists the three answers for each survey ID; Survey 1 answered A, B, C, Survey 2 answered B, A, D and Survey 3 answered C, A, B.

Question 4: Interactive Tableau Dashboard

Question: Develop a Tableau dashboard combining the grouped bar chart, stacked bar chart, and the table for interactive exploration of survey responses.

Aim: To build an interactive Tableau dashboard that combines the grouped bar chart, stacked bar chart and table for exploring survey responses.



Sales Details

Order ID	Product Name	Product Category	Region	Sales Amount (₹)	Quantity	Order Date
S001	Laptop	Electronics	North	60,000	2	01-07-2024
S002	Mobile Phone	Electronics	South	45,000	3	05-07-2024
S003	Headphones	Accessories	East	18,000	4	10-07-2024
S004	T-shirt	Clothing	West	22,000	5	12-07-2024
S005	Refrigerator	Home Appliances	North	30,000	1	15-07-2024
S006	Smartwatch	Accessories	South	12,000	2	18-07-2024
S007	Jeans	Clothing	East	20,000	4	21-07-2024
S008	Microwave Oven	Home Appliances	West	28,000	1	25-07-2024

Result: The dashboard places all three views together, and a question filter updates them so users can explore the responses interactively.

Overall Result

Observation: Out of 9 answers in total, A and B were each chosen 3 times (33.3%), C twice (22.2%) and D once (11.1%).